

Hardware Assembly and Antenna Output Testing

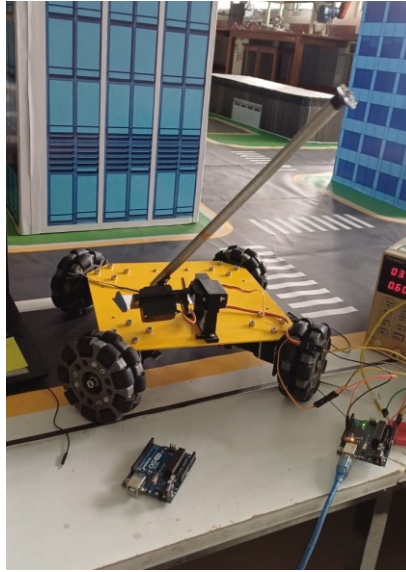


Figure 01 - Hardware Assembling and Testing

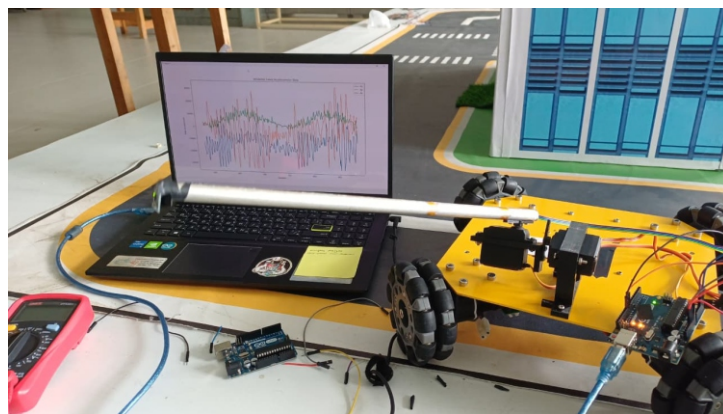


Figure 02 - Hardware Assembling and Testing



Figure 03 - 3D Printed Parts

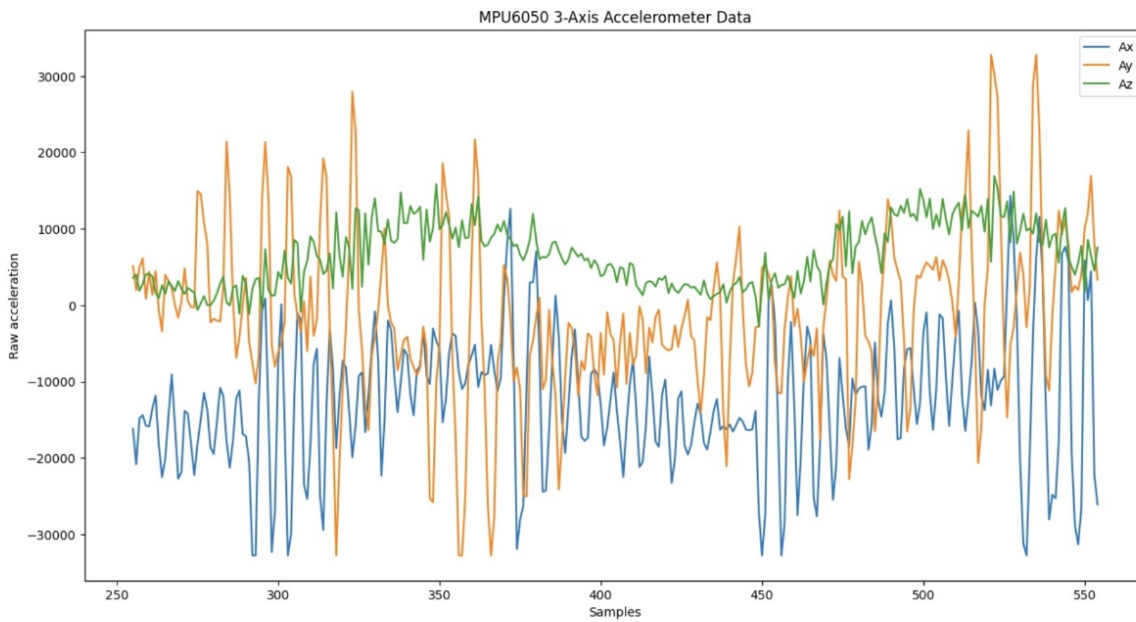


Figure 04 - Tip Accelerometer Sensor Output Waveform

Test code for the implement of sinusoidal motion of the antenna:

```
#INCLUDE <SERVO.H>
#include <WIRE.H>
#include <MPU6050.H>
#include <MATH.H>

SERVO SERVO1;
SERVO SERVO2;
MPU6050 MPU;

FLOAT T = 0;
FLOAT STEPSIZE = 0.02;

VOID SETUP() {
    SERIAL.BEGIN(115200);

    SERVO1.ATTACH(10);
    SERVO2.ATTACH(9);

    WIRE.BEGIN();
    MPU.INITIALIZE();

    DELAY(1000);
}
```

```

VOID LOOP() {
  // SERVO INFINITY MOTION
  INT ANGLE1 = 90 + 50 * SIN(T);
  INT ANGLE2 = 90 + 25 * SIN(2 * T);

  SERVO1.WRITE(ANGLE1);
  SERVO2.WRITE(ANGLE2);

  T += STEPSIZE;

  // MPU6050 READINGS
  INT16_T AX, AY, AZ;
  INT16_T GX, GY, GZ;

  MPU.GETMOTION6(&AX, &AY, &AZ, &GX, &GY, &GZ);

  // SEND TO PYTHON: AX, AY, AZ
  SERIAL.PRINT(AX);
  SERIAL.PRINT(",");
  SERIAL.PRINT(AY);
  SERIAL.PRINT(",");
  SERIAL.PRINTLN(AZ);

  DELAY(15);
}

```